Theorem (Simplified 4.2.1) Let V be a (finite dimensional) completely reducible A-module, with isotypic decomposition $V \simeq \bigoplus_{\lambda \in \hat{A}} \mathrm{Hom}(F_\lambda, V) \otimes F_\lambda$. Let $B = \mathrm{Comm}(A)$.

Then the B-modules $\mathrm{Hom}_A(F_\lambda, V)$ are irreducible.

Proof Set $S := \{\lambda \in \hat{A} : V_{(\lambda)} \neq \{0\}\}$.

Step 1 By the previous lemma, $A \simeq \bigoplus_{\lambda \in S} \mathrm{End}(F_\lambda)$

Step 2 Set $E_\lambda := \mathrm{Hom}(F_\lambda, V)$ so that $V \simeq \bigoplus E_\lambda \otimes F_\lambda$ & $A \simeq \bigoplus_{\lambda \in S} I \otimes \mathrm{End}(F_\lambda)$.

We claim that $B = \bigoplus_{\lambda \in S} \mathrm{End}(E_\lambda) \otimes I$. The inclusion "$\supseteq$" is clear. Next we show "$\subseteq$":

Step 3 Let $b \in B$. By the explicit realization of the B-action, we have $b \cdot (E_\lambda \otimes F_\lambda) \subseteq E_\lambda \otimes F_\lambda$.

Step 4 It is enough to prove that in $\tilde{A} := \mathrm{End}(E \otimes F)$, the commutant of $I \otimes \mathrm{End}(F)$ is $\mathrm{End}(E) \otimes I$: Let $X \in \mathrm{Comm}_{\tilde{A}}(I \otimes \mathrm{End}(F))$. We can write X as

($E \simeq \mathbb{C}^m$, $F = \mathbb{C}^n$): $\quad X = \sum c_{i,j,i',j'} E_{i,j} \otimes E_{i',j'}$. $\quad c_{i,j,i',j'} \in \mathbb{C}$.

$$\left.\begin{aligned} X(1 \otimes E_{r,s}) &= \sum c_{i,j,i',r} E_{i,j} \otimes E_{i',s} \\ (1 \otimes E_{r,s}) X &= \sum c_{i,j,s,j'} E_{i,j} \otimes E_{r,j'} \end{aligned}\right\} \Rightarrow \begin{aligned} &c_{i,j,s,t} = 0 \text{ for } t \neq s \\ &c_{i,j,u,r} = 0 \text{ for } u \neq r \\ &c_{i,j,r,r} = c_{i,j,s,s} \ \forall r,s \end{aligned}$$

From these relations we obtain $X = \sum c_{i,j,r,r} E_{i,j} \otimes I$. Hence $X \in \mathrm{End}(E) \otimes 1$ ∎

Schur-Weyl duality - abstract form.

Let $V := \mathbb{C}^n$. There is a natural representation of $GL_n(\mathbb{C})$ on $V^{\otimes k}$ for every $k \geq 1$:

$$\rho_k : GL_n(\mathbb{C}) \longrightarrow GL(V^{\otimes k})$$
$$T \longmapsto T^{\otimes k} \qquad T^{\otimes k}(v_1 \otimes \cdots \otimes v_k) := Tv_1 \otimes \cdots \otimes Tv_k.$$

There is also an action of S_k on $V^{\otimes k}$: $\sigma_k : S_k \longrightarrow GL(V^{\otimes k})$

$$\sigma_k(s) \cdot v_1 \otimes \cdots \otimes v_k := v_{s^{-1}(1)} \otimes \cdots \otimes v_{s^{-1}(k)}, \ \forall s \in S_k.$$

Lemma The actions of GL_n and S_k mutually commute. That is,

$$\forall g \in GL_n(\mathbb{C}), \forall s \in S_k : \ \rho_k(g)\,\sigma_k(s) = \sigma_k(s)\,\rho_k(g)$$

Theorem 4.2.10 (Schur) Set $\mathcal{A} := \rho_k(GL_n(\mathbb{C}))$ and $\mathcal{B} := \operatorname{Span}(\sigma_k(S_k)) = \sigma_k(\underbrace{\mathbb{C}[S_k]}_{\text{group algebra of } S_k \text{ defined in lecture 1.}})$

Then $\mathcal{B} = \operatorname{Comm}(\mathcal{A})$ and $\mathcal{A} = \operatorname{Comm}(\mathcal{B})$.

Proof **Step 1** Every module for $\mathbb{C}[S_k]$ is completely reducible (this is Maschke's Theorem).

Therefore it is enough to show that $\operatorname{Comm}(\mathcal{B}) \subseteq \mathcal{A}$ (since $\operatorname{Comm}(\mathcal{B}) \supseteq \mathcal{A}$ is evident).

Step 2 Let $\{e_i\}_{i=1}^n$ be the standard basis of $V = \mathbb{C}^n$. For $I = (i_1, \dots, i_k) \in \{1, \dots, n\}^{\times k}$, set

$e_I := e_{i_1} \otimes \cdots \otimes e_{i_k}$. Then $\sigma_k(s)(e_I) = e_{s \cdot I}$ where $s \cdot I := (i_{s^{-1}(1)}, \dots, i_{s^{-1}(k)})$.

Step 3 Any $T \in \operatorname{End}(V^{\otimes k})$ has a matrix $[T] = [a_{I,J}]$:

$$Te_J = \sum_I a_{I,J} e_I.$$

We have $T\sigma_k(s) e_J = \sum a_{I, s \cdot J} e_I$ & $\sigma_k(s) T e_J = \sum a_{I,J} e_{s \cdot I}$

Hence $T \in \operatorname{Comm}(\mathcal{B}) \iff a_{I, sJ} = a_{s^{-1}I, J} \ \forall s \in S_k \ \forall I, J \iff a_{s \cdot I, s \cdot J} = a_{I,J} \ \forall s \in S_k \ \forall I, J$.

Step 4 Set $\langle X, Y \rangle := \operatorname{tr}(XY) \ \forall X, Y \in \operatorname{End}(V^{\otimes k})$. **Claim:** $\langle \cdot, \cdot \rangle|_{\operatorname{Comm}(\mathcal{B}) \times \operatorname{Comm}(\mathcal{B})}$ is nondegenerate.

Proof of Claim: Set $X^{\natural} := \frac{1}{k!} \sum_{s \in S_k} \sigma_k(s) X \sigma_k(s^{-1}) \quad \forall X \in \operatorname{End}(V^{\otimes k})$. ($X \mapsto X^{\natural}$ is projection onto $\mathcal{B}$)

* $X^{\natural} \in \operatorname{Comm}(\mathcal{B})$

* If $Y \in \operatorname{Comm}(\mathcal{B})$ then $\langle X^{\natural}, Y \rangle = \frac{1}{k!} \sum_{s \in S_k} \operatorname{tr}(\sigma_k(s) X \sigma_k(s^{-1}) Y)$

$$= \frac{1}{k!} \sum_{s \in S_k} \operatorname{tr}(\sigma_k(s) X Y \sigma_k(s^{-1})) = \frac{1}{k!} \sum_{s \in S_k} \operatorname{tr}(XY) = \langle X, Y \rangle.$$

* Assume $Y \in \mathcal{B}$ such that $\langle X, Y \rangle = 0 \ \forall X \in \mathcal{B}$

Then by the previous *, $\langle X, Y \rangle = \langle X^{\natural}, Y \rangle = 0$ for all $X \in \operatorname{End}(V^{\otimes k})$,

Hence $Y = 0$.

Step 5 Let $\mathcal{A}^{\perp}$ denotes the orthogonal complement of $\mathcal{A}$ in $\operatorname{Comm}(\mathcal{B})$

Claim $\mathcal{A}^{\perp} = \{0\}$. (This will imply $\operatorname{Comm}(\mathcal{B}) = \mathcal{A}$).

In the rest, we prove the claim.

Step 6 Let $g = [g_{ij}]_{1 \le i,j \le n} \in GL_n(\mathbb{C})$. $\rho_k(g) = g \otimes \cdots \otimes g \Rightarrow [\rho_k(g)] = [g_{I,J}]$ (matrix in basis $\{e_I\}$)

$$g_{I,J} = g_{i_1 j_1} \cdots g_{i_k j_k}$$

Step 7 Let $T \in \mathcal{A}^{\perp}$. Write $T = [a_{I,J}]$ (matrix in $\{e_I\}$)

$$0 = \langle T, \rho_k(g) \rangle = \sum_{I,J} a_{I,J} g_{j_1 i_1} \cdots g_{j_k i_k} \qquad \forall g = [g_{ij}] \in GL_n(\mathbb{C}).$$

Step 8 Set $f_T(X) := \sum a_{I,J} x_{j_1 i_1} \cdots x_{j_k i_k} \qquad X = [x_{ij}] \in M_{n \times n}(\mathbb{C})$

polynomial in n^2 variables.